NE585
NUCLEAR FUEL CYCLES
Front end of the nuclear fuel cycle
3

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Learning objectives

Demonstrating how the nuclear fuel cycle is an holistic system

Explaining why the nuclear fuel cycle is what it is

Apply principle of SWU

Model a nominal solvent extraction system

Model enrichment processes

Demonstrate principles of enrichment and converstion

Some of Chapter 4 in the book

Check out the Pandora's promise movie if it's still on Netflix

There's a lot of pictures here

Learning nodes

Nuclear fuel cycle overview
Locations of current domestic fleet
Plant closures
Nuclear power around the world
Planned construction
Components of the nuclear fuel cycle
Mining
Milling
Conversion
Enrichment
Separative Work Unit (SWU)
Fuel fabrication

Macro reactor operation
CP-1
Hanford B reactor
EBR-I
EBR-II
TREAT
Military use
Reactor concepts
Generations
PWR
BWR
CANDU
Aqueous reprocessing

More learning nodes

Enrichment

History

Processing

Tails

Conversion

Processing

Nuclear fuel cycle overview

Nuclear fuel cycle

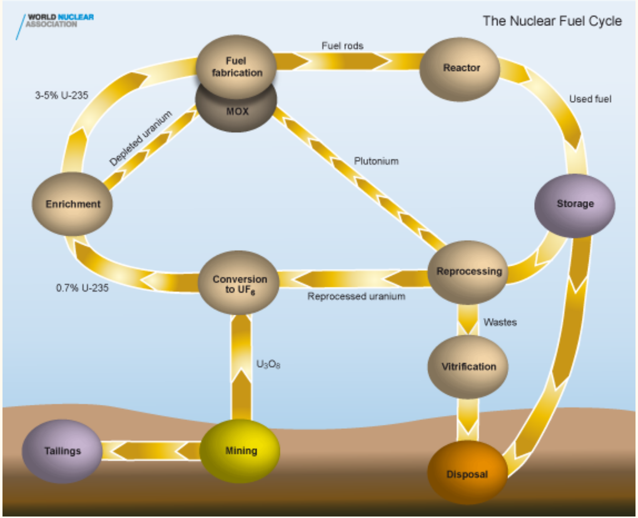


Figure 1. Nuclear fuel cycle

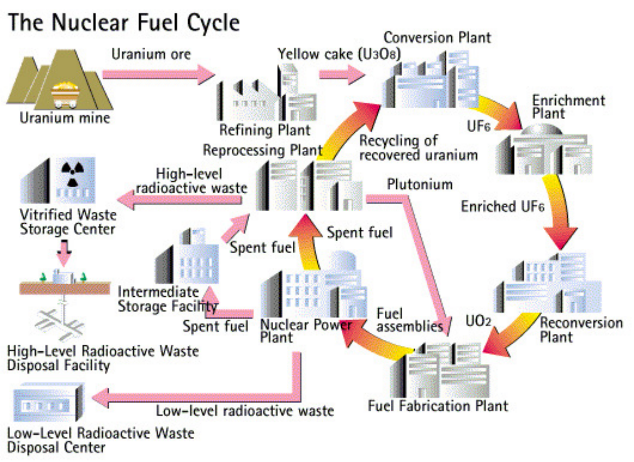


Figure 2. Nuclear fuel cycle

Nuclear fuel cycle

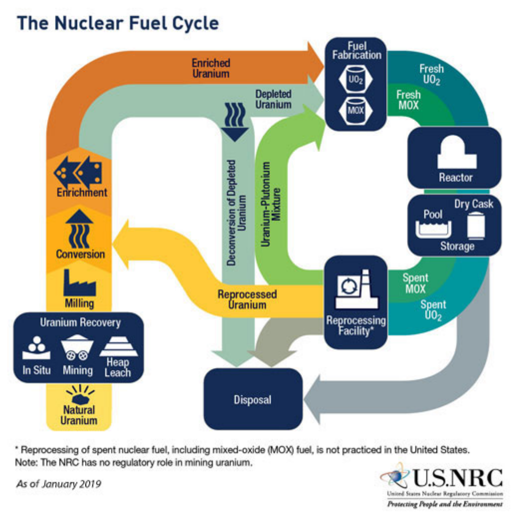


Figure 3. Nuclear fuel cycle

Look at the fuel cycle as systems engineering

Mining

Making the fuel (saw Areva plant in Richland)

What a reactor does and the different kinds

Recycling (what kinds and how)

Back-end management (storage and disposal)

The fuel cycle has a lot of technical but institutional issues too

How are they functionally dependent?

The fuel used in the commercial reactors is UO_2 or $(Pu + U)O_2$

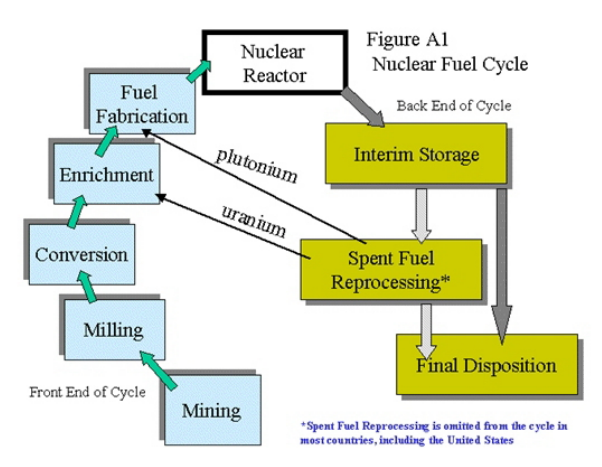


Figure 4. Nuclear fuel cycle

Front end – Making the fuel and putting it in the reactor

Back end – What happens when you take it out of the reactor

Locations of current domestic fleet

Nuclear Power Plant sites

Figure 5. Nuclear Power Plant sites

Currently operating NPPs

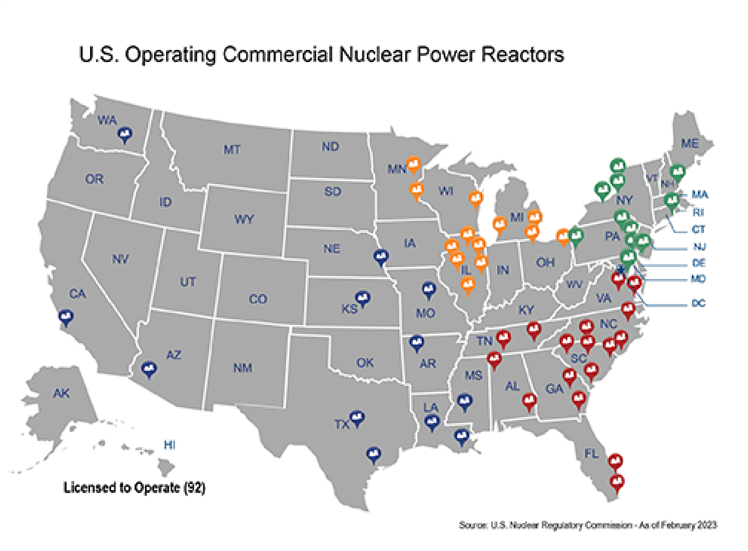


Figure 6. Currently operating NPPs

NRC list of power reactor units

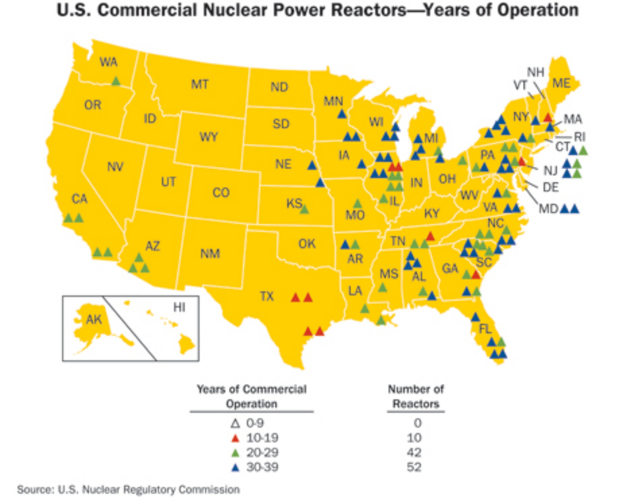


Figure 7. NPP years of operation

Plant closures list on the OER

Recent plant closures

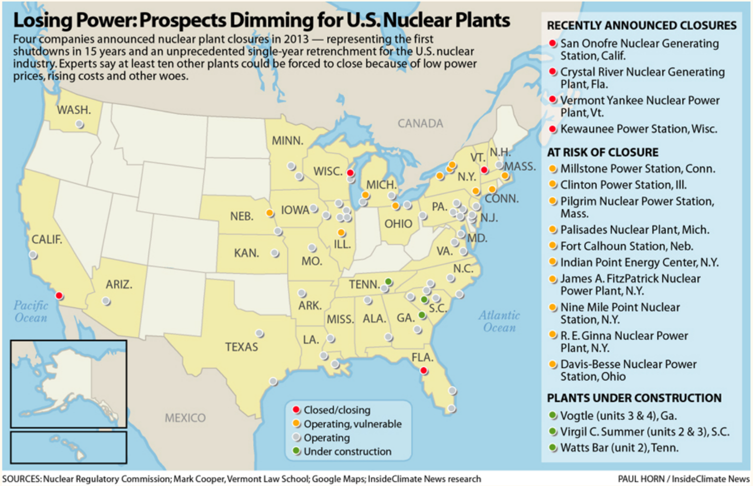


Figure 8. Recent plant closures

Let’s look at other nations

NPPs in Europe

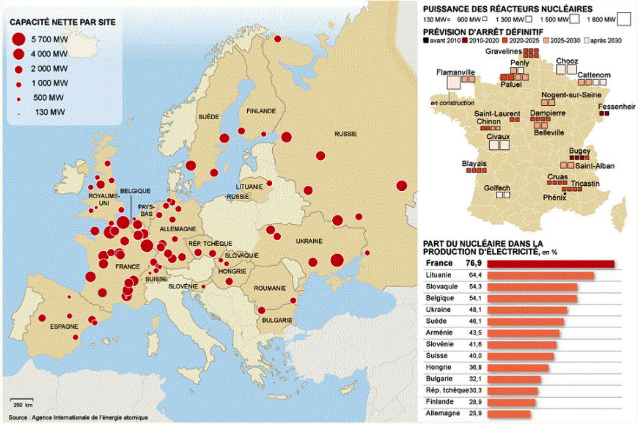


Figure 9. NPPs in Europe

NPPs in Japan

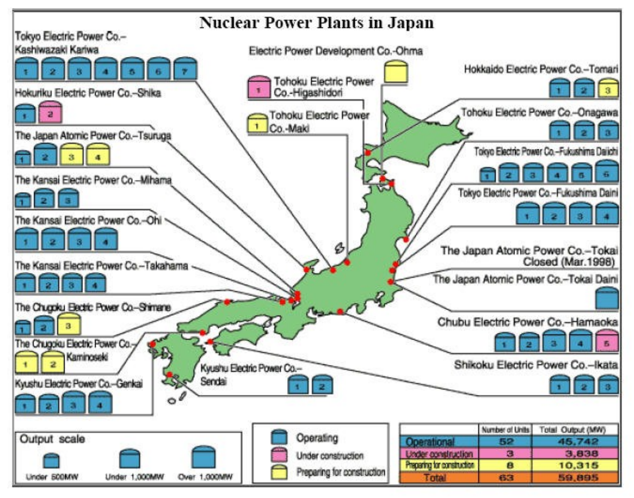


Figure 10. NPPs in Japan

Where are they going to dispose of everything?

Figure 11. NPPs in Japan

Same problem with Korea

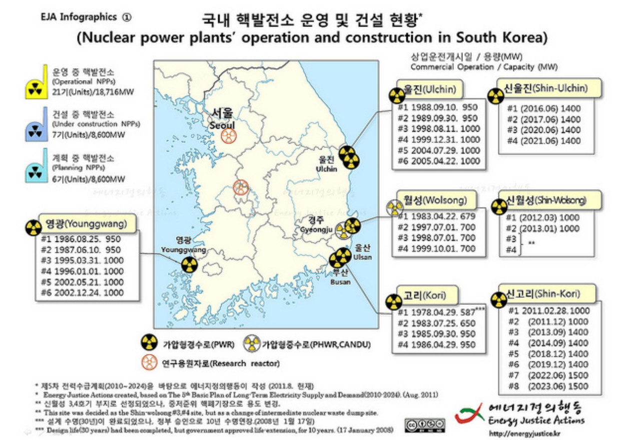


Figure 12. NPPs in Korea

NPPs in Asia

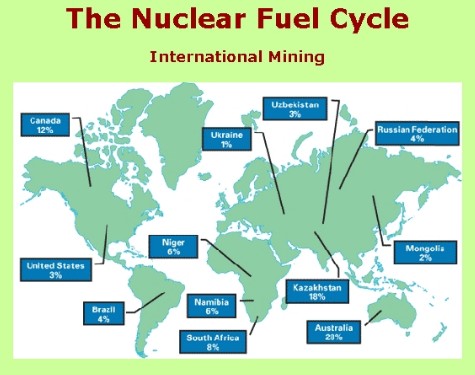
Figure 13. NPPs in Asia

Plans For New Reactors Worldwide from the World Nuclear Association

Components of the nuclear fuel cycle

Mining

We have to find uranium first



U_3O_8 Nuclear Fuel Price Indicators

Extract the ore from the ground

What do you do if you don't have it?

Check out Australia

Milling

Uranium ore is refined by milling

The ore contains UO_2, UO_3 and other heavy metal oxides

Usually using solvents to leach it from the ground

Then it is chemically treated to obtain 'yellowcake'

Lots of contaminated scrap is produced though (mill tailings)

Usually dumped as sludge in ponds (lots)

Radioisotopes (radon), heavy metals, chemicals, etc.

We have plenty of uranium resources

But what if we decide it is too harmful to mine?

Conversion

U_3O_8 is converted to UF_6 prior to enrichment

Mined U contains ^{234}U, ^{235}U, ^{238}U, .0054%, .72%

Because it is a gas at not so high temperature which is easier for enrichment

Enrichment

UF_6 gas is then enriched with ^{235}U

Enriched to 3% to 5%

The byproduct is depleted uranium DU

Used for armor and bullets

Any idea why?

Enrichment uses centrifuge cascasdes

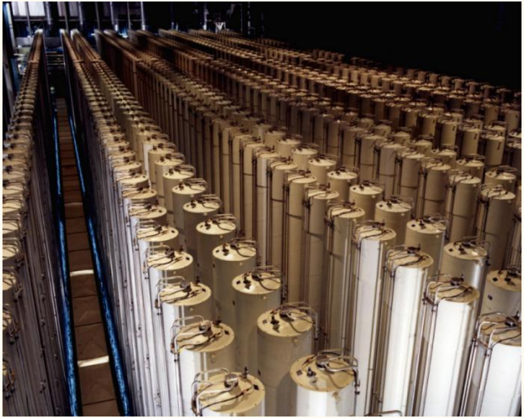


Figure 14. Centrifuge cascades

Enrichment takes advantage of thermodynamics

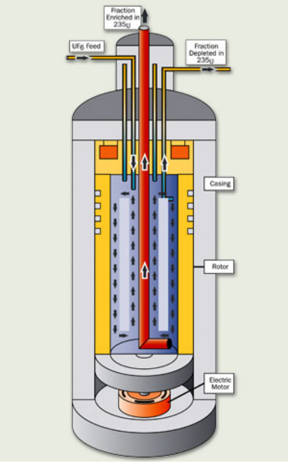


Figure 15. Centrifuge process

Separative Work Unit (SWU)

SWU is the to enrich uranium from natural abundance to reactor/weapon grade

Dumb acronyms are why people hate nuclear

OER documents on SWU

$$x_F = 0.0072 \quad (1)$$

$$x_P = 0.03 - 0.05 \quad (2)$$

$$x_W = 0.0002 - 0.0003 \quad (3)$$

Amount of feed for target product increases linearly with x_P by mass conservation

Enrichment is just based on mass conservation

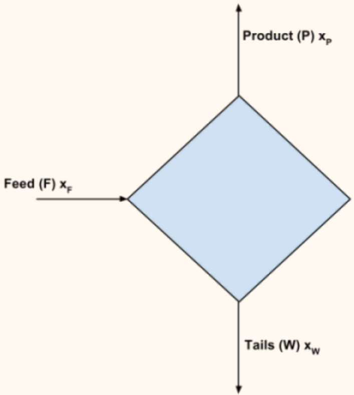


Figure 16. Mass conservation

$$F = P + W \tag{4}$$

$$F \cdot x_F = P \cdot x_P + W \cdot x_W \tag{5}$$

SWU measures thermodynamic work to separate isotopes

SWU has units of kg because of course it does

Separation potential is a measure of differences in Gibbs free energy

$$SWU \equiv [P \cdot V(x_P) + W \cdot V(x_W)] - F \cdot V(x_F) \tag{6}$$

$$V(x_i) \equiv (2x_i - 1) \cdot ln\frac{x_i}{1 - x_i} \tag{7}$$

$$\frac{SWU}{P} = V(x_P) + \frac{W}{P}V(x_W) - \frac{F}{P}V(x_F) \tag{8}$$

The derivation is based on entropy changes

Change in entropy in binary mixture of gases (paper in OER)

What stands out?

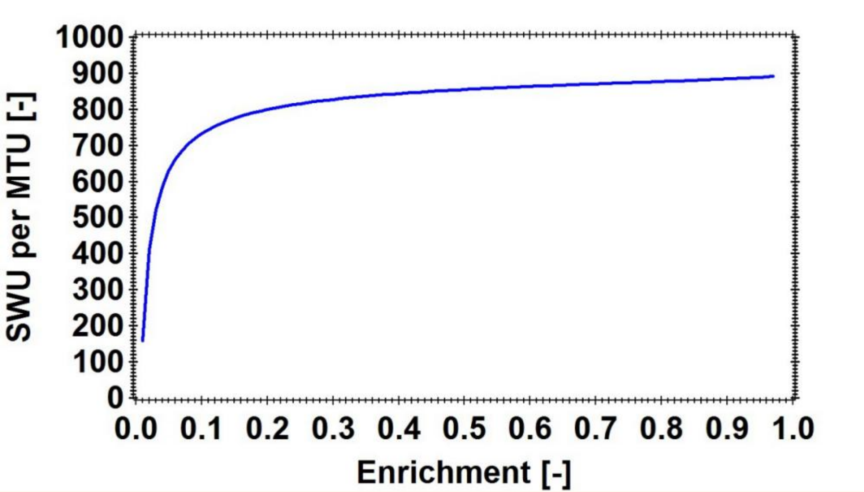


Figure 17. SWU curve varies with enrichment

Fuel Fabrication

The fuel fabrication process (LWRs) makes ceramic UO_2 pellets

Why ceramic?

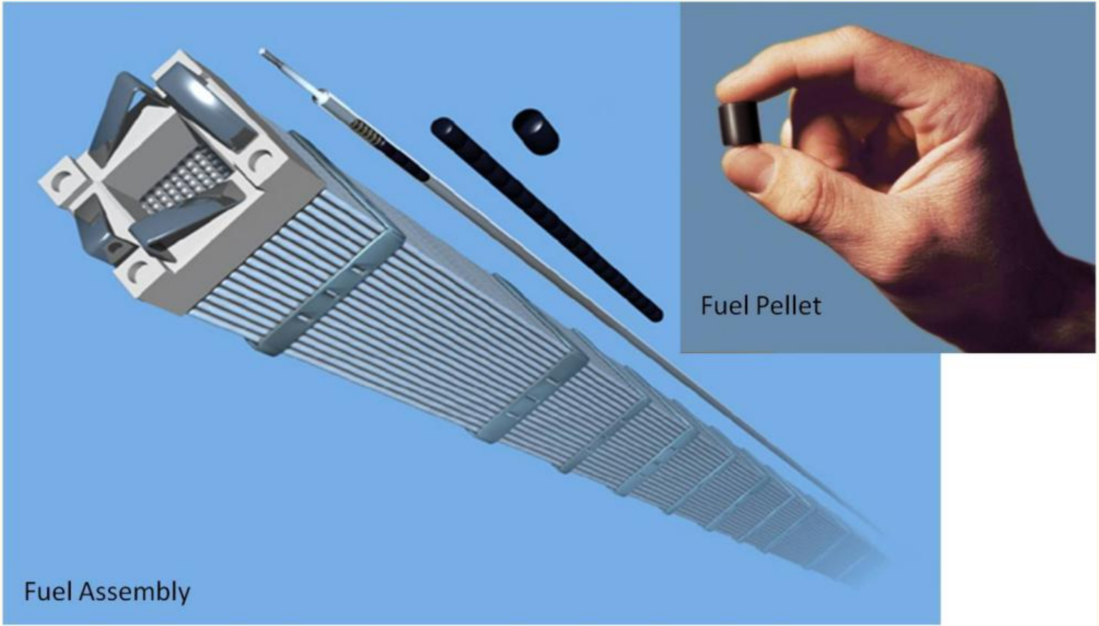
Is there anything better?

Fuel rods contain a Zr-based alloy

Paper on generating hydrogen gas

$$Zr + 2H_2O \rightarrow ZrO_2 + 2H_2 \tag{9}$$

That's kind of what happened at Fukushima (article on Nature)



Fuel Pellet
Fuel Assembly

How much energy do you get from one atom of uranium versus carbon?

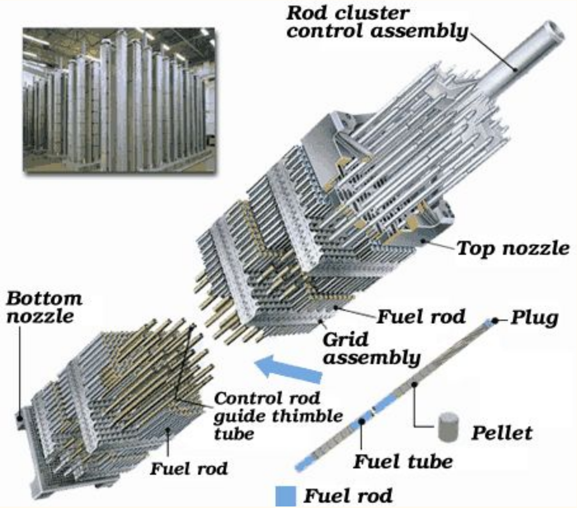


Figure 18. Uranium fuel pellet and assembly

Assemblies are about 12 feet tall

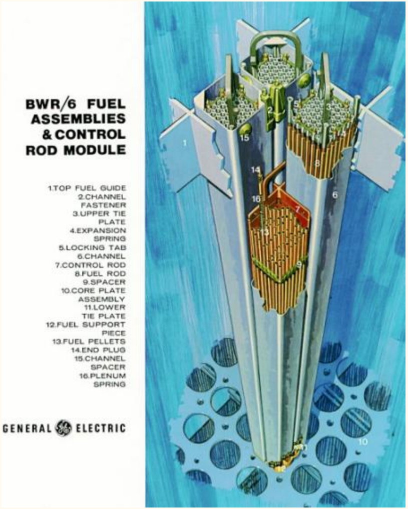


Figure 19. BWR assembly

Macro reactor operation

Chicago Pile-1 (CP-1)

The first man made reactor was constructed at the University of Chicago in 1942

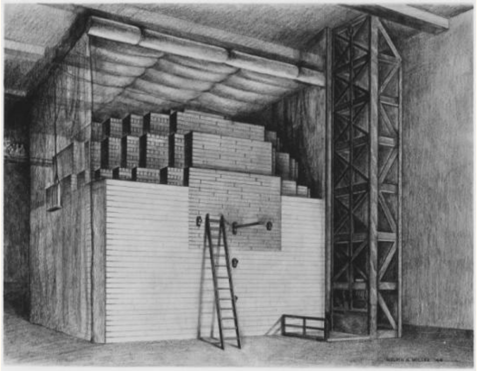


Figure 20. Chicago Pile-1

University of Chicago videos on the OER

Made of uranium and graphite blocks

Cadmium coated rods

And I have a piece of the graphite

Let’s look at pictures of the Hanford B reactor

Same design as CP1

Hanford B Reactor

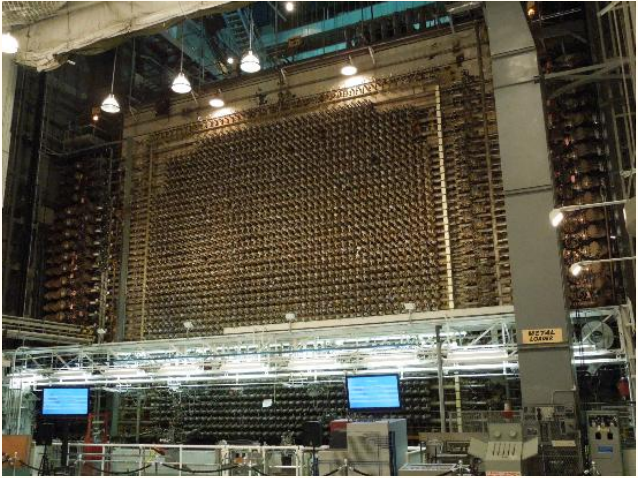


Figure 21. Hanford B Reactor

Hanford B Reactor

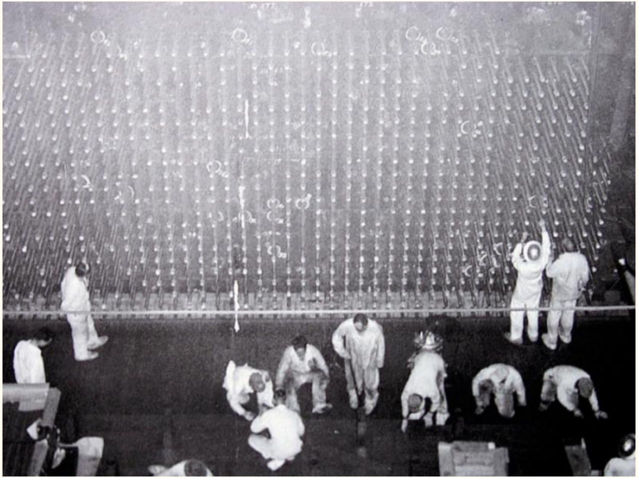


Figure 22. Hanford B Reactor

Hanford B Reactor

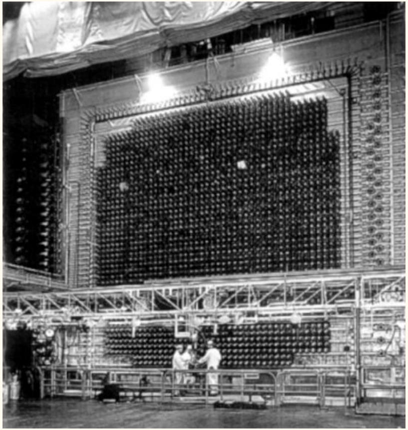


Figure 23. Hanford B Reactor

Hanford B Reactor

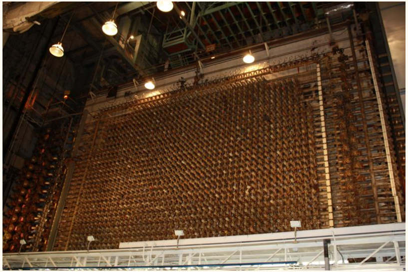


Figure 24. Hanford B Reactor

Hanford B Reactor

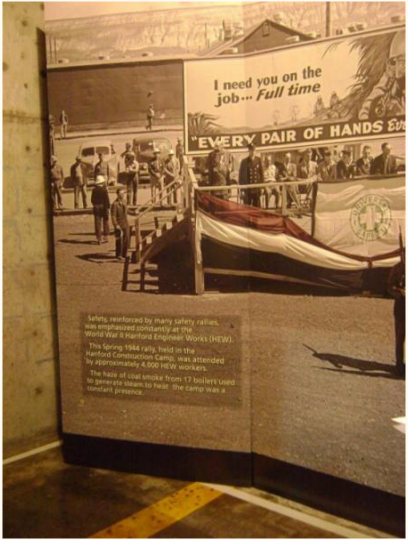


Figure 25. Hanford B Reactor

Hanford B Reactor

EXTRA The Villager EXTRA

PEACE!

OUR BOMB CLINCHED IT!

Plant Will Not Close!

Japs Surrender

Message from W. O. Simon

Food & Drug Stores Close Tomorrow

Only Recreational And Service Facilities Open

"Our Bomb" Does the Job

"It's Peace"

Report Last Nite Awakens Village

Figure 26. Hanford B Reactor

Hanford B Reactor

Figure 27. Hanford B Reactor

Hanford B Reactor

Figure 28. Hanford B Reactor

Hanford B Reactor

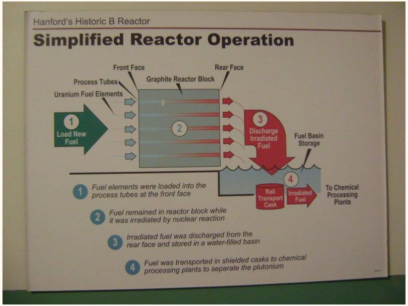


Figure 29. Hanford B Reactor

Hanford B Reactor

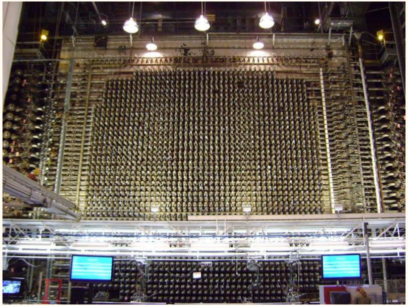


Figure 30. Hanford B Reactor

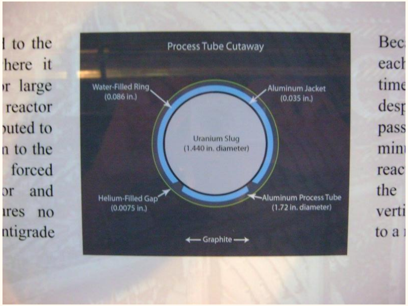


Figure 31. Hanford B Reactor

Hanford B Reactor

Figure 32. Hanford B Reactor

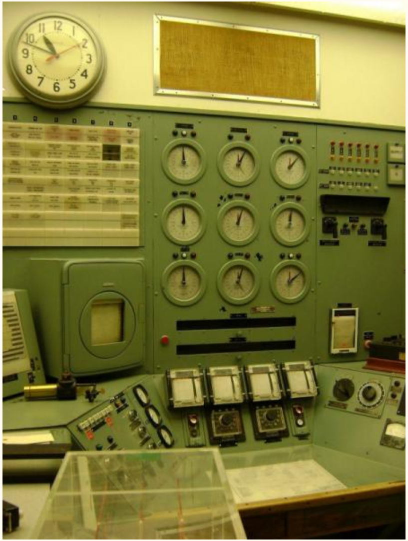


Figure 33. Hanford B Reactor

Figure 34. Hanford B Reactor

Hanford B Reactor

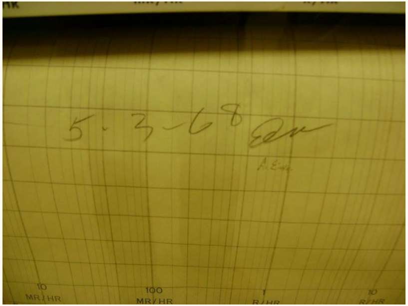


Figure 35. Hanford B Reactor

Natural reactor

A natural reactor actually happened about 2 billion years ago

Fuel cycle on the OER

Neutronics on the OER

16 sites operated for about 10^5 years at 100 kW thermal

French company found samples of 0.60% of ^{235}U

How much is there normally?

Uranium rich mineral deposits were infiltrated by groundwater which served as moderator

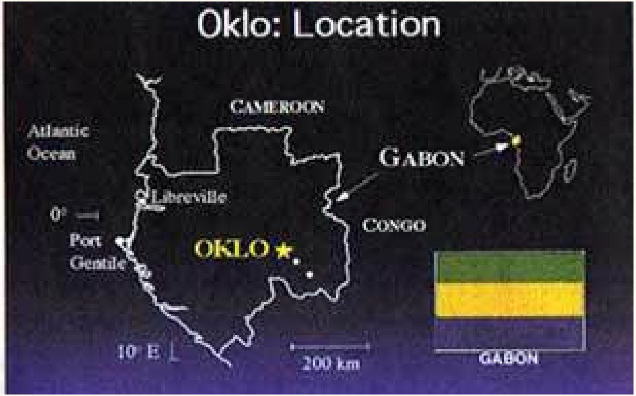


Figure 36. Oklo location

How the Oklo reactors worked

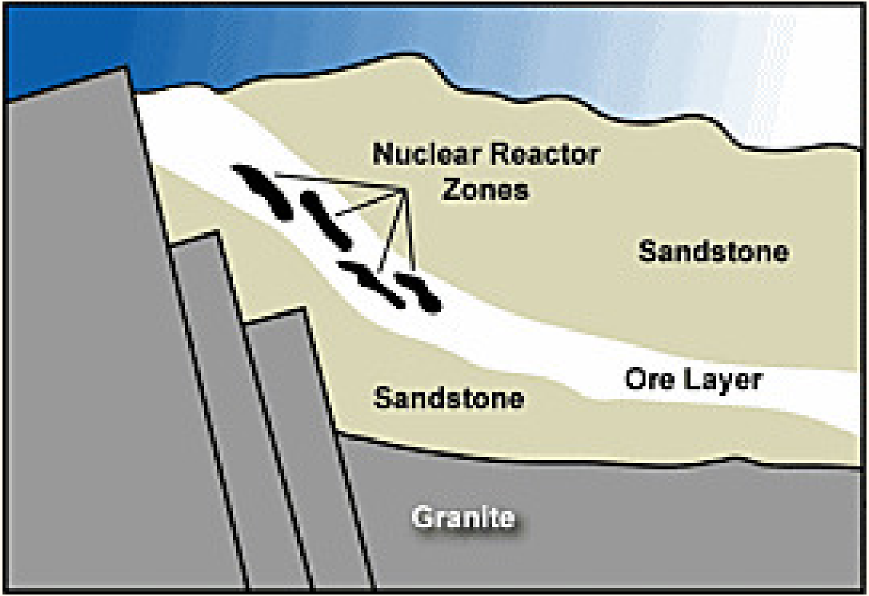


Figure 37. How the Oklo reactors worked

EBR-I

EBR-I was the first reactor to generate electricity

In 1951 – now a museum, open to the public for free

Location of EBR-I

About 200 kW of electricity

Also was a breeder and first to confirm Fermi's theory

'Fast' reactor, metal fuel and molten salt designs

Then the commercial reactors switched to ceramic and water

Why?

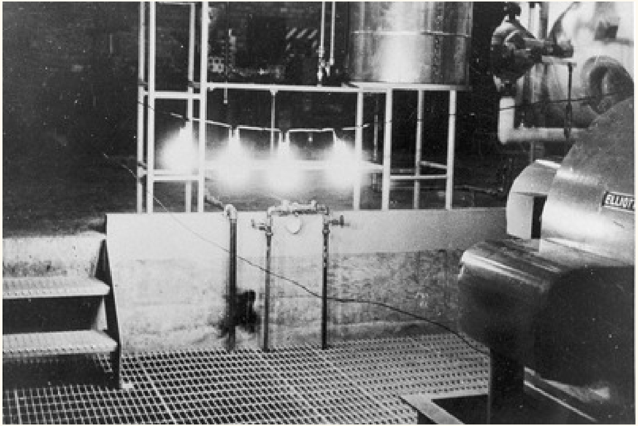


Figure 38. EBR-I lit up 4 light bulbs

More pictures

EBR-I museum

Figure 39. EBR-I museum

EBR-I museum

Figure 40. EBR-I museum

EBR-I museum

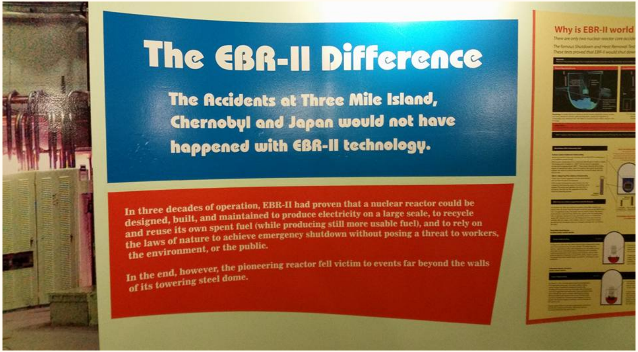


Figure 41. EBR-I museum

EBR-I museum

Figure 42. EBR-I museum

EBR-I museum

Figure 43. EBR-I museum

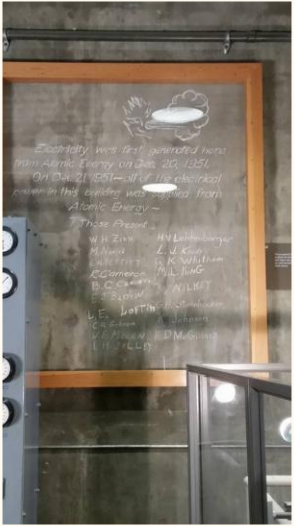


Figure 44. EBR-I museum

Figure 45. EBR-I museum

Figure 46. EBR-I museum

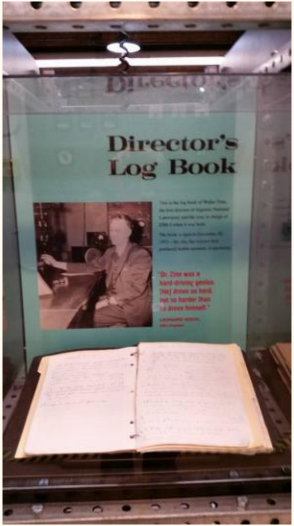


Figure 47. EBR-I museum

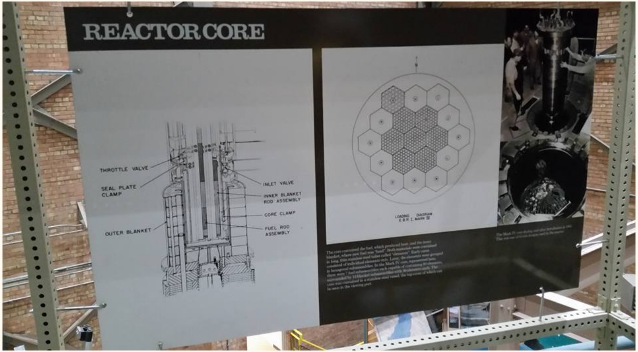


Figure 48. EBR-I museum

EBR-I museum

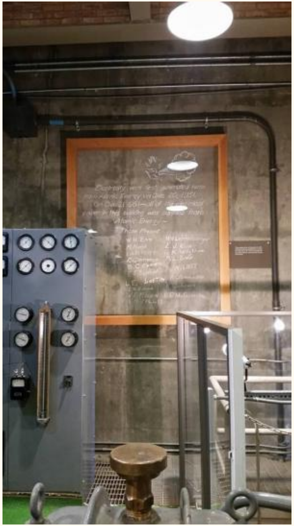


Figure 49. EBR-I museum

EBR-I museum

Figure 50. EBR-I museum

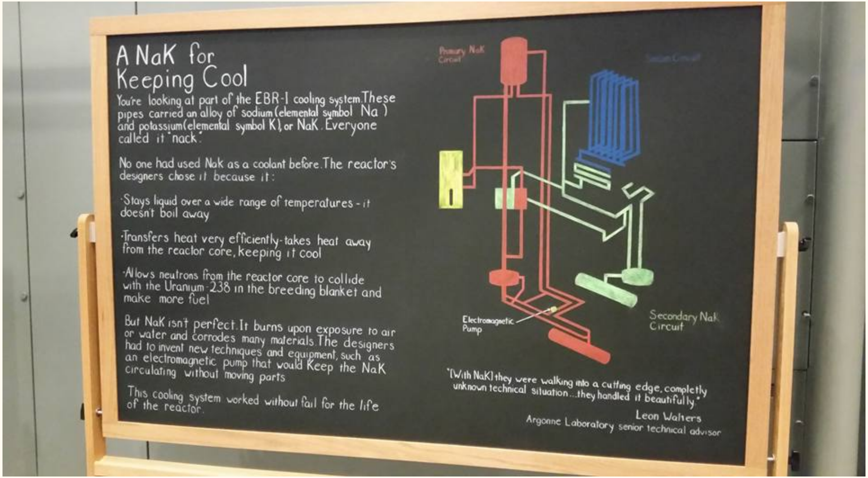


Figure 51. EBR-I museum

Figure 52. EBR-I museum

EBR-II

EBR-II is an Sodium Fast Reactor (SFR) design for 62.5 MWth that operated from 1964 to 1994

19 MWe electricity operated for 30 years with no accidents

Engineering scale facility

Breeder reactor with onsite reprocessing (pyroprocessing)

Famous test of passive safety systems video

EBR-II schematics

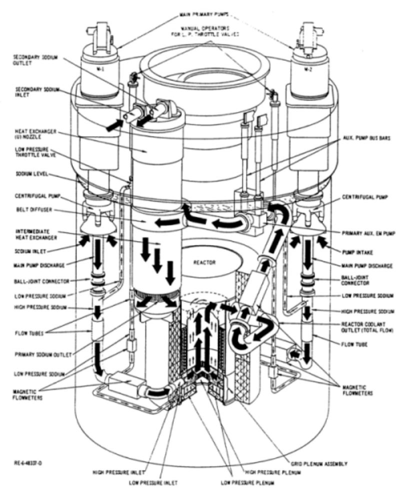


Figure 53. EBR-II schematics

TREAT

TREAT is an air cooled, graphite moderated thermal reactor

Operation from 1959 to 1994

Transient reactor tests to simulate all sorts of reactor accidents

George Imel

Restarted and working on experiments!

Because of Fukushima, TREAT accident tolerant fuels

Everyone is excited

100 kW; transients up to 19 GW

TREAT in person

Figure 54. TREAT

Military use

Nuclear reactors are used on submarines and aircraft carriers

Newsreels on the Nautilus launch in 1954

25 year life span

Nimitz-class aircraft carriers have 2 reactors, about 100 MWe, 23 year life span

How about now?

Nautilus submarine

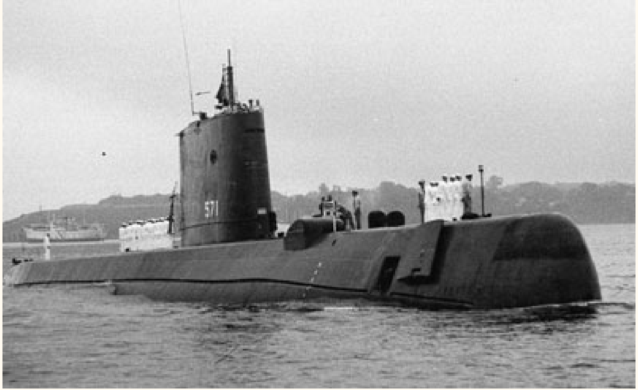


Figure 55. Nautilus submarine

Aircraft carriers have reactors too

Figure 56. Aircraft carrier enters the danger zone. Top Gun theme classic.

Reactor concepts

Generations

Reactors were first built in the 1950s

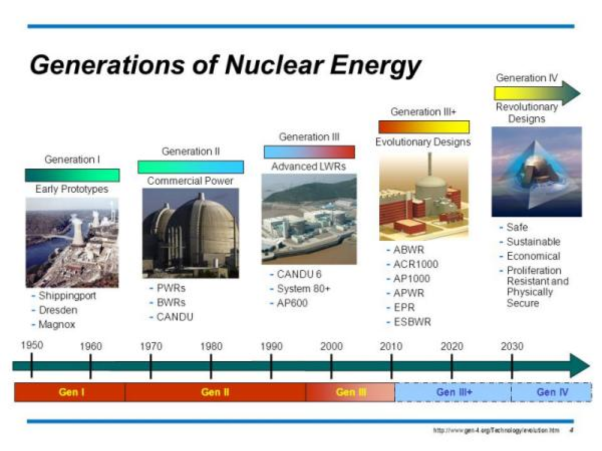


Figure 57. Reactor generations

Gen IV is essentially based on Gen I

Different material for fuels and coolant

Moderators not needed for some

Passive safety systems

Proliferation issues – so they say

PWR

The PWR heats water but does not boil

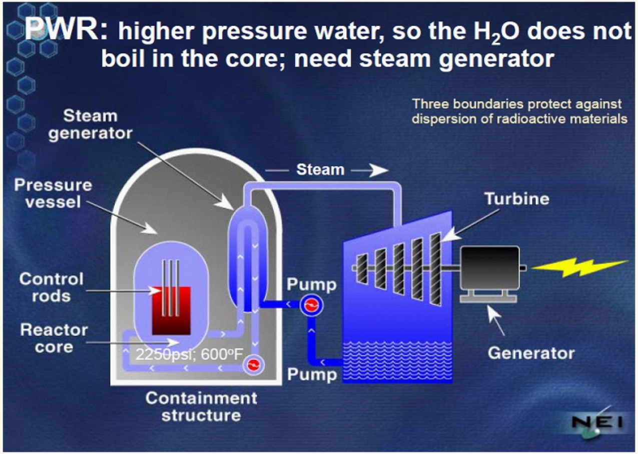


Figure 58. PWR

BWR

The BWR does boil water in the reactor vessel

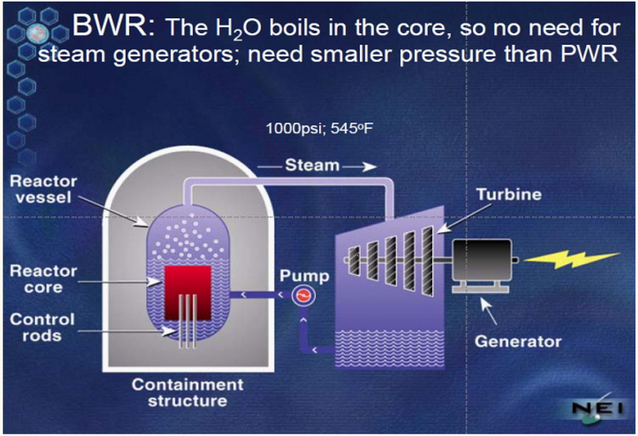


Figure 59. BWR

CANDU

The **CANDU** design uses heavy water (moderator) and natural uranium (also thermal)

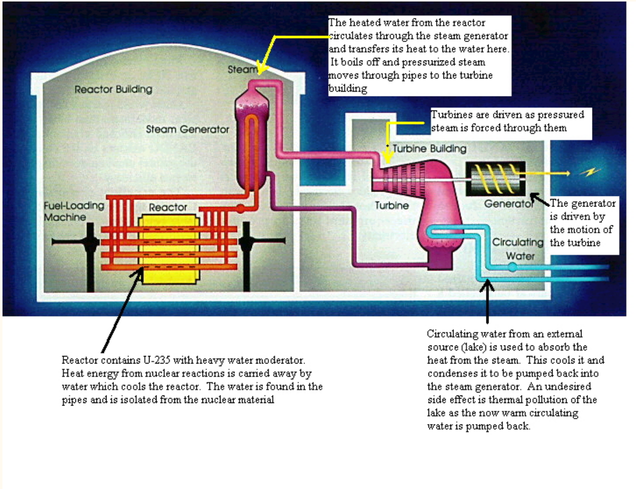


Figure 60. CANDU

More on CANDU

Used in Canada and marketed to other nations

South Korea, China, India have some

PWR design (two loops)

The heavy water is less absorbing so natural U can be used

But also more collisions would be needed than in LWRs

The core would be bigger

But can use used fuel from LWRs and a Th cycle

So is there an advantage? Technical? Institutional?

The **CANDU** design uses heavy water (moderator) and natural uranium (also thermal)

Figure 61. Homer Simpson is bad at his job

Aqueous reprocessing

Aqueous recycling is the commercially available process for LWRs

More information in the OER

Terry Todd leading expert

Major nations are France, Japan, Russia, India

'Recycle' is better for the brand

About 1% 235U and 1% Pu in used fuel

The most common is PUREX

This uses an organic phosphate and nitric acid

Solvent extraction process where U and Pu are separated

Then created into MOX fuel

So for PUREX what is a big risk?

PUREX flowsheet

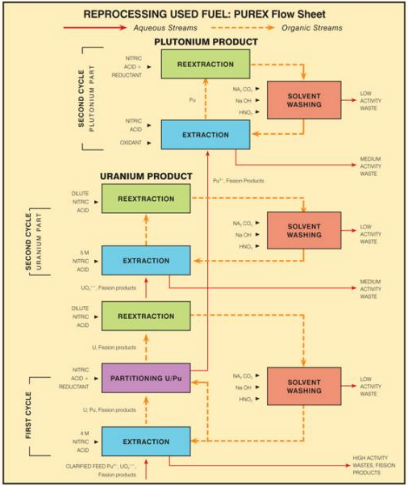


Figure 62. PUREX flowsheet

U and Pu are chemically separated through organic solvent equilibrium extraction

U and Pu exist in a number of valence states

Because of different redox potentials, reduction or oxidation can be done to one without disturbing the other

Each exhibits different solubilities in organic solvent TBP

(1) Chop up everything, remove FP gases, dissolve in aqueous solution

(2) Separate U/Pu from fission products by dissolving in solvent

(3) Reduce Pu to 3+ because it is highly insoluble in solvent (U 6+)

(4) Back extract Pu into aqueous solution

(5) Strip U from solvent into aqueous solution

U and Pu are chemically separated through organic solvent equilibrium extraction

Aqueous solution is nitric acid

Does generate a lot of liquid waste

Mass transfer to solvent phase – extraction

Mass transfer to aqueous phase –“ stripping

Mass transfer operates in a continuous, multistage, countercurrent mode

High separation factor

High processing rates

Streams flow countercurrently, so concentration is relatively constant

Solvent must be recycled because it's not cheap

First applied to extract Pu for weapons production in 1952 at Hanford site

Metals in 4+ and 6+ are extracted

$$(UO_2)^{2+}(aq) + 2(NO_3)^{-}(aq) + 2TPB(o) \leftrightarrow UO_2(NO_3)_2 \cdot 2TBP(o) \tag{10}$$

$$An^{4+}(aq) + 4NO_3^{-}(aq) + 2TBP(o) \leftrightarrow An_2(NO_3)_4 \cdot 2TBP(o) \tag{11}$$

Pu (as 4+) co-extracted with U

Anything 3+ or lower is not

Waste is vitrified into a borosilicate glass waste form for disposal

Well characterized material with 50+ years of scientific study

Very stable

PUREX mass conservation

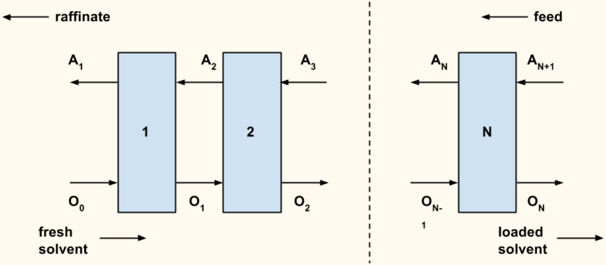


Figure 63. PUREX stage extraction

Typical mass conservation is used to model solvent extraction

This is from Benedict

Distribution coefficient = organic/aqueous phase concentration at equilibrium

$$D \equiv \frac{y}{x} \tag{12}$$

When an aqueous solution of an extractable component is brought into equilibrium with an immiscible solvent for the component and the two phases are then separated, the component will be found distributed between the two phases

We then derive the material balance on the extractable component

$$Fz = Fx + Ey \tag{13}$$

With D for the system defined before

PUREX single stage extraction

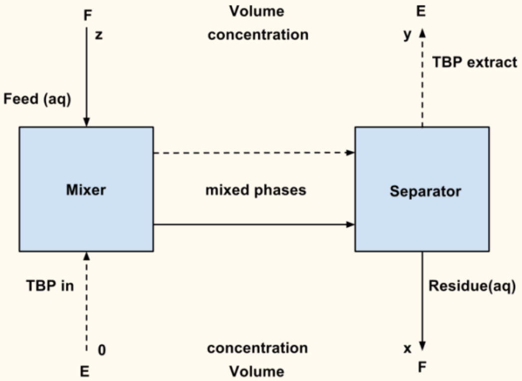


Figure 64. PUREX single stage extraction

Derive the concentration that is extracted

$$D \equiv \frac{y}{x} \tag{14}$$

$$Fz = Fx + Ey \tag{15}$$

$$\therefore y = \frac{Dz}{1 + \frac{ED}{F}} \tag{16}$$

And the fraction extracted is –

$$\rho = \frac{Ey}{Fz} = \frac{1}{1 + \frac{F}{ED}} \tag{17}$$

For design then, you only need 3 parameters

Derive the solvent feed needed for a given extraction fraction

$$\frac{E}{F} = \frac{\rho}{D(1-\rho)} \tag{18}$$

What is the functional dependence of $E(\rho)$?

Amount of used fuel isn't actually a variable

Reduce the amount of solvent by using multiple contact stages

With many stages, high extraction fraction can be obtained

Solvent is expensive, so recycling stages are also added to strip out extract

Multiple components can be extracted from organic phase if D is different enough

So this would be needed to completely separate U, Pu

Physical conditions affecting D –

- Element extracted
- Redox potential of aqueous phase
- Solvent
- Concentrations
- $[H^+]_{aq}$

PUREX flowsheet

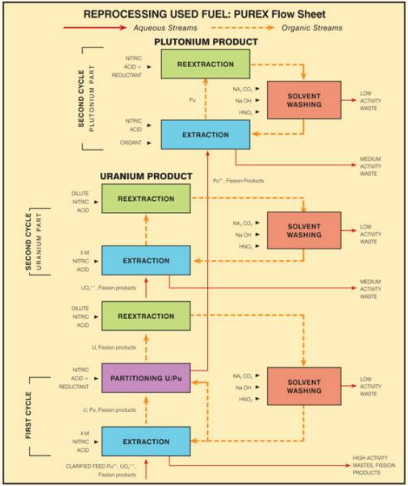


Figure 65. PUREX flowsheet

Multistage extraction

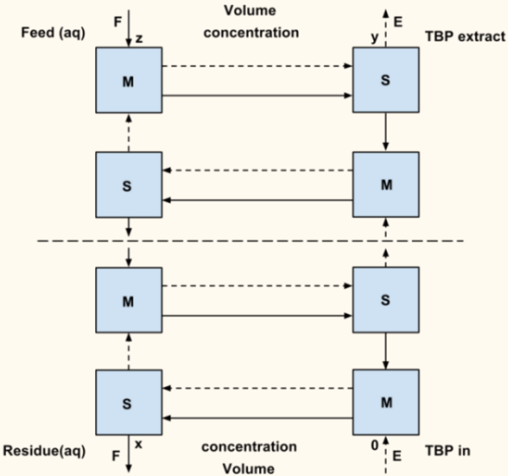


Figure 66. Multistage extraction

Multistage extraction engineering diagram

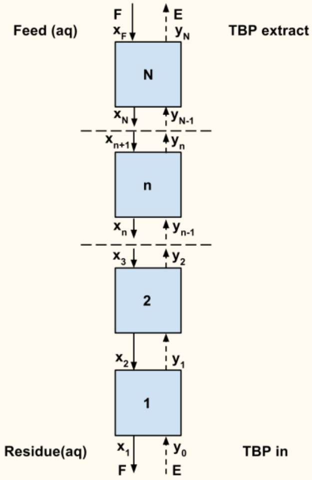


Figure 67. Multistage extraction engineering diagram

Now apply same mass conservation to multistage extraction

Assume equilibrium between phases

Material balance on extractable component below stage n –

$$Ey_0 + Fx_n = Ey_{n-1} + Fx_1 \tag{19}$$

$$y_{n-1} - y_0 = \frac{F}{E}(x_n - x_1) \tag{20}$$

$\therefore$ for any stage –

$$y_n \equiv D_n x_n \tag{21}$$

McCabe-Thiele diagram can be used for constructing graphical solution for for stage concentrations

Basically gives the number of stages for an extraction concentration x_F

By graphing $y_{n-1} - y_0 = \frac{F}{E}(x_n - x_1)$ on y versus x

Construct operating line to satisfy the above and pass through x_1, y_0 with slope $\frac{F}{E}$ relative to equilibrium line $y_n = D_n x_n$

Cannot really change equilibrium line because it is based on D

Reduce the concentration of the extractable component from x_F to x_1 with $\frac{F}{E}$

Start at x_1, y_0 and move up

McCabe-Thiele diagram

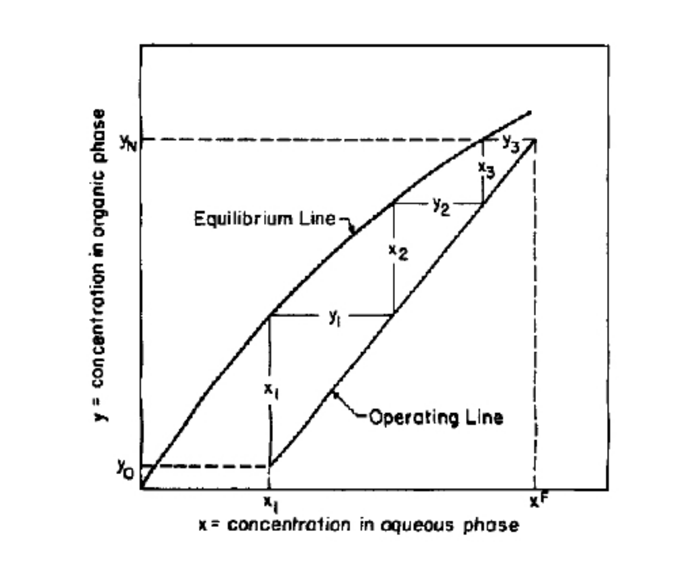


Figure 68. McCabe-Thiele diagram

Derive the extraction factor for multiple stages

$$\rho \equiv \frac{Ey_N}{Fx_F} \tag{22}$$

Or –

$$1 - \rho = \frac{x_1}{x_F} \tag{23}$$

What does this tell us?

And feed ratio can be defined in terms of $f(\rho)$ as –

$$\frac{E}{F} = \frac{\rho x_F}{y_N - y_0} \tag{24}$$

How many stages for complete extraction?

What does that look like on McCabe-Thiele?

Apply these concepts to model multistage extraction based on Kremser equation course notes

$$y_{n-1} - y_0 = \frac{F}{E}(x_n - x_1) \tag{25}$$

$$y_n \equiv D_n x_n \tag{26}$$

$$\beta \equiv \frac{DE}{F} \tag{27}$$

Then –

$$y_n = \beta(y_{n-1} - y_0) + Dx_1 \tag{28}$$

$$y_N = \frac{\beta^N - 1}{\beta - 1}(Dx_1 - y_0) + y_0 \tag{29}$$

This can also be applied if there are two extractable components in the feed

$$D_A \; D_B \; \beta_A \; \beta_B \tag{30}$$

Define decontamination factor. What does this mean?

$$f_{AB} \equiv \frac{\rho_A}{\rho_B} \tag{31}$$

$$f_{AB} = \frac{\beta_A}{\beta_B}\left(\frac{\beta_A^N - 1}{\beta_B^N - 1}\right)\left(\frac{\beta_B^{N+1} - 1}{\beta_A^{N+1} - 1}\right), \quad y_0 = 0 \tag{32}$$

Conversion & Enrichment

Uranium enrichment derived from World War II

Needed to make bombs

Then used the uranium for submarines

When moving to commercial power, infrastructure already in place

Gaseous diffusion first invented at Oak Ridge

Uranium hexafluoride used for enrichment

Yellowcake needs to be purified first

Impurities are rare earths, chlorine cadmium

Uranium readily forms chemical compounds (lots of oxidation states)

Can be extracted by organic solvents

Uranium forms complexes that can be precipitated out

There are different forms that are mined

$$
\begin{aligned}
&xUO_2 \cdot yUO_3 \\
&0 < y/x < 2
\end{aligned} \tag{33}
$$

$$K_2O \cdot 2UO_3 \cdot V_2O_5 \cdot xH_2O \tag{34}$$

$$CaO \cdot 2UO_3 \cdot P_2O_5 \cdot xH_2O \tag{35}$$

Goal is to get it as UO_3^{2+} and then acid or alkaline leaching

$$UO_3 + 2H^+ \rightarrow UO_2^{2+} + H_2O \tag{36}$$

$$UO_2 + 2Fe^{3+} \rightarrow UO_2^{2+} + 2Fe^{2+} \tag{37}$$

Acid leaching is like PUREX using organic solvent

$$UO_2^{2+} + 2NO_3 + 2TBP \rightarrow UO_2(NO_3)_2 \cdot 2TBP \tag{38}$$

$$UO_2(NO_3)_2 + 2H_2O_2 + H_2O \rightarrow UO_4 \cdot 2H_2O + 2HNO_3 \tag{39}$$

Precipitates uranium peroxide

$$2Na_2S_2O_3 + UO_4 + H_2O \rightarrow Na_2S_4O_6 + UO_3 + 2NaOH \tag{40}$$

Or use ammonium hydroxide

Acid leaching is like **PUREX** using organic solvent

$$UO_2^{2+} + 6NH_4(OH) \rightarrow (NH_4)_2U_2O_7 + 4NH_4^+ + 3H_2O \tag{41}$$

Then dry and calcine to get UO_3

If alkaline leaching is used, recover U_3O_8

Uranium hexafluoride has favorable properties

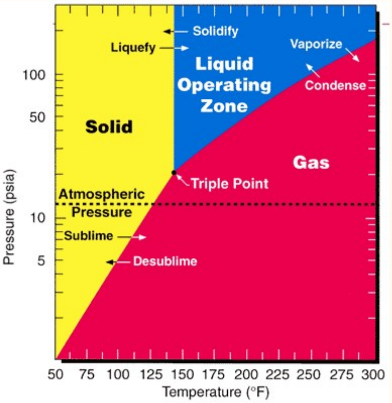


Figure 69. Uranium hexafluoride phase diagram

Solid at room temperature

Easy to handle

Relatively low pressure

Not too high temperature to sublimate

Conversion applies hydrofluor process

Crushed U_3O_8 and UO_3 is reduced by hydrogen to get 'crude' UO_2

$$U_3O_8 + 2H_2 \rightarrow 3UO_2 + 2H_2O \tag{42}$$

$$UO_3 + H_2 \rightarrow UO_2 + H_2O \tag{43}$$

Conversion applies hydrofluor process

Then –

$$UO_2 + 4HF \rightarrow UF_4 + 2H_2O \tag{44}$$

UF_4 is a solid green salt with a high melting point (1700^oF)

Treated with fluorine gas and distilled –

$$UF_4 + F_2 \rightarrow UF_6 \tag{45}$$

Enrichment based on different diffusion rates

Separation factor –

$$\alpha \equiv \frac{v_5}{v_8} = \sqrt{\frac{m_8}{m_5}} \tag{46}$$

Based on kinetic energy of the molecule $kT = \frac{1}{2}mv^2$

This just says lighter molecules move faster

$\alpha = 1.004289$ for uranium enrichment

Higher value, easier enrichment

This isn't that high

Low enrichment takes a lot of energy

Cascades are designed to enrich

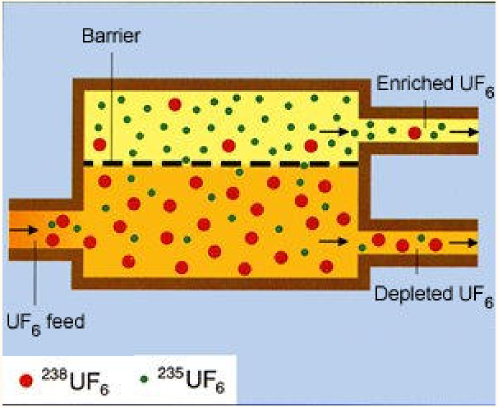


Figure 70. Centrifuge diffusion

Basic operation is the stage

Join stages to form cells

Cells form unit

Together all that is a cascade

Enriching and stripping stages added for efficiency

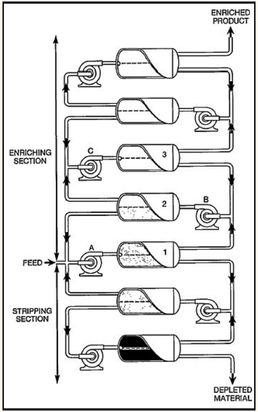


Figure 71. Gas diffusion cascade

Feed enters to A-1

Enriched feed goes through B-2

Depleted through the bottom stage

Moving up is higher enrichment

Enrichment based on mass conservation

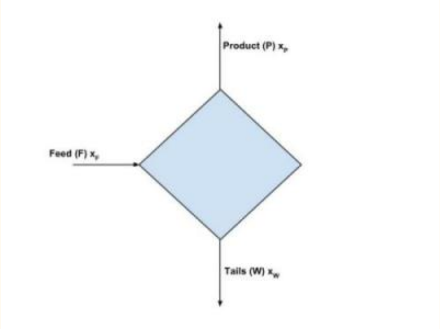


Figure 72. Mass conservation

$$F = P + W \tag{47}$$

$$x_F F = x_P P + x_W W \tag{48}$$

Separation factor ratio between product and tails –

$$\alpha \equiv \frac{x_P(1 - x_P)^{-1}}{x_W(1 - x_W)^{-1}} \tag{49}$$

And mass flow

x_F = weight fraction ^{235}U feed into system

x_P = weight fraction ^{235}U product (enrichment target)

x_W = weight fraction ^{235}U waste stream (depleted U)

F = mass flow rate feed

P = mass flow rate product

W = mass flow rate waste stream

Derive feed factor

x_F fixed at 0.00711

x_P enrichment target

x_w 0.002 - 0.003 depends on company

Feed factor gives mass of U needed to feed into system for target enrichment –

$$\frac{F}{P} = \frac{x_P - x_W}{x_F - x_w} \tag{50}$$

Derive feed factor

Similarly waste factor –

$$\frac{W}{P} = \frac{x_P - x_F}{x_F - x_W} = \frac{F}{P} - 1 \tag{51}$$

How are tails managed?

Only one company enriches by gaseous diffusion

1992 – United States Enrichment Corporation established under Energy Policy Act

1998 – privatized with so you can buy stock

Operates plant in Paducah up to 5%

Enrichment also done by centrifuge

1970 – URENCO established with United Kingdom, Dutch, Germany

1977 – Construction of US plant in Ohio, but Reagan scrapped it in favor of laser

Rotating drum causes centrifugal force to compress gas molecules to the outer wall

Velocity of molecules separates differing those of differing weights

Light molecules collect near center of drum

Still based on conservation of mass

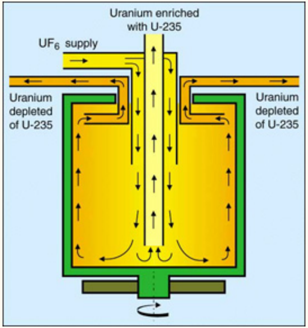


Figure 73. Gas centrifuge

Lighter U^{235} molecules collect toward center

Rotational motion of he centrifuge accelerates gas molecules toward wall

Counteracting diffusive force thermal motion of molecules seeks to distribute the gas evenly

Two forces balance to create a dynamic equilibrium

Pressure distribution in the rotor that is function of the molecular mass

$$\alpha = exp^{\frac{(m_8 - m_5)\omega^2 a^2}{2RT}} \approx 1 + \frac{(m_8 - m_5)\omega^2 a^2}{2RT} \quad (52)$$

$$p(r) = p_0 exp^{\frac{m\omega^2 r^2}{2RT}} \quad (53)$$

Design based on separative power

$$SWU = \frac{\pi \rho D L}{2} \left(\frac{\Delta m \omega^2}{2RT} \right)^2 \tag{54}$$

density of uranium hexafluoride; isotope self-diffusion; mass difference; length of rotor

Proportional to length and 4th power of speed

Inverse to temperature squared

What does this say about how to design the centrifuge?

It's actually small SWU per centrifuge, so up to 100,000 needed forr 9 million SWU/y

That would be needed for industrial scale

There are only 3 companies in the USA interested in centrifuges

USEC Inc. (USEC)

Louisiana Energy Services (LES)

AREVA Enrichment Services (AES)

USEC submitted application to operate a commercial facility

American Centrifuge Plant (ACP) August 2004

NRC issued a license April 2007

Construction of the facility is currently on hold (did not get DOE grant)

Chapter 11 in 2014 – now called Centrus Energy Corp

Louisiana Energy Services activity

LES submitted license application NRC December 2003 for a 3 million SWU/year facility

One-third of the United States demand

Lea County, New Mexico

June 2006 – NRC issued LES a license to construct and operate

Began operations in June 2010 while continuing construction

Part of URENCO

July 2014 – All cascades operational

AREVA Enrichment Services (AES) activity

December 2008 – AES submitted a license application for Eagle Rock Enrichment Facility to NRC

18 miles west of Idaho Falls!

September 2010 – NRC issued the Safety Evaluation Report (NUREG-1951)

February 2011 – Final Environmental Impact Statement (NUREG-1945)

October 2011 – NRC issued AES license to construct and operate the Eagle Rock Enrichment Facility

Construction currently inactive status update

Lasers can be used for enrichment

Figure 74. Shark with laser beam

Is laser enrichment any good?

Started in 1985

Absorption spectrum of uranium metal vapor

More than 300,000 lines at visible wavelengths

Sufficient displacement between lines

Selective excitation of ^{235}U based on laser wavelength

Requires liquid uranium at $2300^o C$

Laser at 5915 angstroms to excite ^{235}U

Ultraviolet light at 3100 angstroms ionizes ^{235}U

Then it is collected in a Faraday cup

Low throughput, high temperature, high energy

Current developments

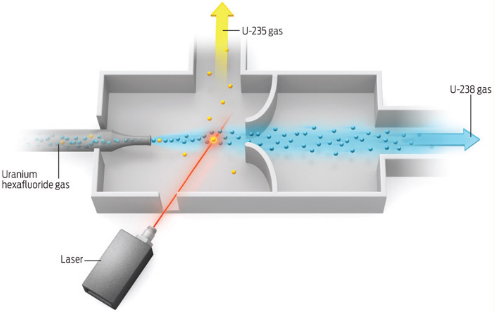


Figure 75. Laser enrichment

GE got an operating license

Doesn't seem like much progress, limited testing with DU in Paducah

Different process than what we were just talking about though

Looking to construct in Wilmington

Han shot first!

Figure 76. Han Solo shoots Greedo

Molecular separation uses uranium hexafluoride

Uses different vibrational and rotational states between isotopes

Almost the same procedure

^{235}U molecule is dissociated and precipitates to a $^{235}UF_5$ powder

Need to reduce the vapor pressure to 77K

Underrated 80s gem

Figure 77. Real Genius

SILEX seems to be state of the art

Uses carbon dioxide laser at 10.8 um and amplified to 16 um

Infrared spectrum

Preferentially excites $^{235}UF_5$

Again trapped electromagnetically

Still inefficient due to laser frequency

Price used to be set by government

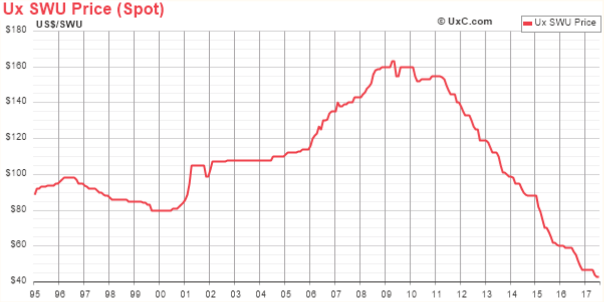


Figure 78. SWU price

From Atomic Energy Act 1954

Prices then set by USEC based on market conditions

Used to be high because of high energy requirements of gaseous diffusion plants

What now?

World capacity is estimated at 60 million SWU

World enrichment capacity

US demand at 15 million SWU per year

Import a lot from Europe, Russia, China

What's the problem here?

Acronyms I

BWR Boiling Water Reactor.

CANDU Canada Deuterium Uranium Reactor.

CP-1 Chicago Pile-1.

DOE United States Department of Energy.

EBR-I Experimental Breeder Reactor I.

EBR-II Experimental Breeder Reactor II.

FP Fission Product.

LWR Light Water Reactor.

MOX Mixed Oxide.

MWe Megawatts-electric.

MWth Megawatts-thermal.

Acronyms II

NPP Nuclear Power Plant.

NRC Nuclear Regulatory Commission.

OER Online Educational Resource.

PUREX Plutonium Uranium Reduction Extraction.

PWR Pressurized Water Reactor.

SFR Sodium Fast Reactor.

SWU Separative Work Unit.

TBP Tributyl Phosphate.

TREAT Transient Reactor Test Facility.

UI University of Idaho.

UIIF Idaho Falls Center for Higher Education.